17/9/26

EXP 13.

IMPLEMENT YOUR OWN PING PROGRAM

AIM: To implement a ping program using python raw sockets to check the reachability of host.

CODE:

```
import socket, struct, time, os

def checksum(data):
    total = 0
    for i in range(0, len(data)-1, 2):
        total += (data[i] << 8) + data[i+1]

    if len(data) % 2:
        total += data[-1] << 8
    total = (total >> 16) + (total & 0xffff)
    total += total >> 16
    return (~total)

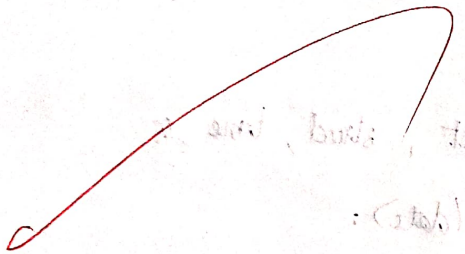
def ping(host):
    try:
        dest_addr = socket.gethostbyname(host)
        sock = socket.socket(socket.AF_INET,
                             socket.SOCK_RAW,
                             socket.IPPROTO_ICMP)

        packet_id = os.getpid() & 0xffff
        sequence = 1
        header = struct.pack("!BBHHH",
                             8,
                             0,
```

Sample Input / Output -

Enter destination host : 8.8.8.8

Reply from 8.8.8.8 : time = 24.53 ms



```
data = b"PING"
checksum = checksum(header + data)
header = struct.pack("!BBHHH", 8, 0, 0, 0,
                     packet_id, sequence)

packet = header + data
start = time.time()
sock.sendto(packet, dest_addr, 1))
sock.settimeout(2)
sock.recvfrom(1024)
rtt = (end - start) * 1000
print("Reply from", dest_addr, ": time = %.2f ms" % rtt)
sock.close()

except sock.timeout:
    print("Request timed out")

except PermissionError:
    print("Permission denied. Run the program as administrator
          (sudo).")
    host = input("Enter destination host")
    ping(host)
```

RESULT

Thus, the ping program was successfully implemented